AS · Unit 2 · WPH12 Waves & Electricity

Current, P.D. & Resistance

Every circuit is about charges moving and carrying energy. This page builds the language for it: current, potential difference, resistance and power. Then you'll see why resistivities of different materials range over more than thirty powers of ten, why a filament lamp's graph curves, and why a thermistor's resistance falls when it warms up while a metal's rises.

Electric current

An electric current is a flow of **charged particles**, called **charge carriers**. In a metal they are **free electrons** (also called conduction or delocalised electrons) that are not tied to any one atom. In a salt solution or a molten salt they are **ions**, positive ones moving one way and negative ones the other.

DEFINITION

Electric current is the rate of flow of charged particles: $I = \dfrac{\Delta Q}{\Delta t}$. A current of 1 ampere is a flow of 1 coulomb of charge per second ($1\ \text{A} = 1\ \text{C}\ \text{s}^{-1}$).

Current

FORMULA SHEET

$$I = \frac{\Delta Q}{\Delta t}$$

I: current / A ΔQ: charge passing a point / C Δt: time taken / s

- Charge comes in whole numbers of the electron charge $e = 1.60 \times 10^{-19}\ \text{C}$. A current of 1 A means $1 \div 1.60 \times 10^{-19} = 6.25 \times 10^{18}$ electrons pass every point each second.
- **Conventional current** is drawn from the positive terminal round the circuit to the negative terminal. Electrons actually drift the other way. The direction was chosen before anyone knew about electrons, and it makes no difference to any calculation.
- Rearranged, $\Delta Q = I\Delta t$, so the **area under a current–time graph is the charge** that has flowed.

COMMON MISTAKE

Current is **not used up** by a lamp or a motor. Every coulomb that goes in comes out the other side, because charge is conserved. What the lamp takes is **energy** from the charges. That's the next idea: potential

difference.

WORKED EXAMPLE

A phone charger delivers a steady current of 2.0 A for 45 minutes. Calculate the charge that flows and the number of electrons this represents.

Show solution

Convert the time to seconds first: $45 \times 60 = 2700$ s.

$$\Delta Q = I\Delta t = 2.0 \times 2700 = 5400 \text{ C}$$

$$N = \frac{\Delta Q}{e} = \frac{5400}{1.60 \times 10^{-19}} = 3.4 \times 10^{22} \text{ electrons}$$

Potential difference

Think of each coulomb of charge as a delivery van. The cell loads every coulomb with energy, and the components in the circuit unload it: a lamp turns it into light and thermal energy, a motor into kinetic energy. The **potential difference** across a component tells you how much energy each coulomb drops off there.

DEFINITION

The **potential difference (p.d.)** between two points is the energy transferred (work done) per unit charge moving between them: $V = \frac{W}{Q}$. One volt is one joule per coulomb ($1 \text{ V} = 1 \text{ J C}^{-1}$).

Potential difference

FORMULA SHEET

$$V = \frac{W}{Q}$$

V: p.d. / V W: energy transferred / J Q: charge / C

A voltmeter measures the p.d. **between two points**, so it is always connected **in parallel** across a component. An ammeter measures the current **through** a component, so it goes **in series**. An ideal voltmeter has infinite resistance (it takes no current) and an ideal ammeter has zero resistance (it takes no p.d.).

WORKED EXAMPLE

15 C of charge flows through a lamp and transfers 180 J of energy. Calculate the p.d. across the lamp and the energy transferred by each electron.

Show solution

$$V = \frac{W}{Q} = \frac{180}{15} = 12 \text{ V}$$

Each electron carries charge e, so it transfers $W = eV$:

$$W = 1.60 \times 10^{-19} \times 12 = 1.92 \times 10^{-18} \text{ J } (= 12 \text{ eV})$$

Drift velocity: I = nqvA

The free electrons in a metal are always rushing about randomly at around 10^6 m s^{-1}, but in every direction equally, so on average they go nowhere. Connect a cell and an electric field pushes them all slightly one way. On top of their random motion they gain a slow average **drift velocity** along the wire. That drift is the current.

Deriving I = nqvA by counting charges

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Take a wire of cross-sectional area A. It contains n charge carriers per unit volume (the **number density**, in m^{-3}), each with charge q, all drifting at speed v.

2. Pick a cross-section of the wire and watch it for a time Δt. Each carrier moves a distance $v\Delta t$, so every carrier that starts within $v\Delta t$ of the cross-section gets through. They fill a cylinder of volume:

$$\text{volume} = A \times v\Delta t$$

3. The number of carriers in that cylinder, and the charge they carry, are:

$$N = nAv\Delta t \qquad \Delta Q = Nq = nAv\Delta t\, q$$

4. Current is charge per unit time, so divide by Δt:

$$I = \frac{\Delta Q}{\Delta t} = \frac{nAv\Delta t\, q}{\Delta t} \quad \Rightarrow \quad \boxed{I = nqvA}$$

Current and drift velocity

FORMULA SHEET

$$I = nqvA$$

n: number of charge carriers per m³ / m^{-3} q: charge on each carrier / C v: drift velocity / m s^{-1} A: cross-sectional area / m²

WORKED EXAMPLE

A copper wire of cross-sectional area 1.0 mm² carries a current of 2.0 A. Copper has 8.5×10^{28} free electrons per m³. Calculate the drift velocity of the electrons.

Show solution

Convert the area: $1.0\ \text{mm}^2 = 1.0 \times 10^{-6}\ \text{m}^2$. Rearrange $I = nqvA$:

$$v = \frac{I}{nqA} = \frac{2.0}{8.5 \times 10^{28} \times 1.60 \times 10^{-19} \times 1.0 \times 10^{-6}} = 1.5 \times 10^{-4}\ \text{m s}^{-1}$$

That is 0.15 mm per second. An electron would take nearly two hours to drift one metre.

Why resistivities range so widely

The biggest difference between materials is n, the number of charge carriers per unit volume:

Type of material	Example	Free charge carriers n
Metal	copper	$\approx 10^{29}\ \text{m}^{-3}$ (about one per atom)
Semiconductor	germanium, silicon	$\approx 10^{16}$ to $10^{19}\ \text{m}^{-3}$ at room temperature
Insulator	glass, plastic	almost none at room temperature

Put the same p.d. across the same-sized pieces of each. By $I = nqvA$, a material with far fewer charge carriers carries a far smaller current, so its resistance, and its resistivity, is far larger. The drift velocity for a given p.d. depends on the material too, but n varies by so many powers of ten that it dominates.

EXAM TIP

Two wires of different thickness in series carry the **same current**, so by $I = nqvA$ the drift velocity is **larger in the thinner wire** ($v \propto 1/A$ for the same n and I). Say "same current because charge is conserved" before you use the equation.

Resistance and Ohm's law

As free electrons drift through a metal they keep colliding with the vibrating positive ions of the lattice, handing over energy (which is why wires get warm). **Resistance** measures how hard it is to push a current through a component.

DEFINITION

The **resistance** of a component is the p.d. across it divided by the current through it: $R = \frac{V}{I}$. One ohm is one volt per ampere ($1\ \Omega = 1\ \text{V}\ \text{A}^{-1}$).

Ohm's law: for a metallic conductor at constant temperature, the current is directly proportional to the potential difference across it ($I \propto V$).

Resistance

FORMULA SHEET

$$R = \frac{V}{I}$$

R: resistance / Ω V: p.d. across the component / V I: current through it / A

KEY IDEA

A definition and a law are different things

$R = V/I$ **defines** resistance. It works for any component at any instant, even a diode. **Ohm's law** is an experimental result that only some components obey: if $I \propto V$, then V/I stays the same and the component is **ohmic**. Ohm's law is the special case where the resistance doesn't change with the current, which needs the temperature (and other physical conditions) to stay constant.

Current–p.d. characteristics

To find a component's characteristic, put it in series with an ammeter, connect a voltmeter across it, and vary the p.d. with a variable supply or a potential divider. Reverse the connections to get negative values. Plot current I against p.d. V. Here are the four you need to sketch, recognise and explain.

Component	Shape of the I–V graph	Why
Ohmic conductor (fixed resistor, metal wire at constant temperature)	Straight line through the origin, the same in both directions	$I \propto V$: the resistance is constant
Filament lamp	S-shaped curve: steep near the origin, flattening off at higher p.d. in both directions	A bigger current heats the filament; the lattice ions vibrate more, the electrons collide more often, and the resistance **increases**
NTC thermistor	Curve that gets **steeper** as the p.d. increases, in both directions	A bigger current heats it; more conduction electrons are released, so the resistance **decreases**
Diode	Almost zero current in reverse; in the forward direction almost zero until about 0.6 V, then the current rises very steeply	It conducts in one direction only: very high resistance in reverse, low resistance once forward-biased above about 0.6 V

COMMON MISTAKE

The resistance at a point on an I–V graph is V/I at **that point**. It is **not** 1/(gradient of the tangent). The two are only equal for an ohmic conductor, where the graph is a straight line through the origin. Also check which way round the axes are: on an I–V graph a steeper line means a *smaller* resistance.

WORKED EXAMPLE

A filament lamp takes a current of 0.15 A at 2.0 V and 0.25 A at 6.0 V. Calculate its resistance at each p.d. and explain the difference.

Show solution

$$R_{2\,\mathrm{V}} = \frac{2.0}{0.15} = 13\ \Omega \qquad R_{6\,\mathrm{V}} = \frac{6.0}{0.25} = 24\ \Omega$$

At 6.0 V the current is larger, so more energy is transferred to the filament every second and it is much hotter. The metal ions vibrate with larger amplitude, the free electrons collide with them more often, and the drift velocity for a given p.d. is lower. So the resistance is higher.

Electrical power and energy

Power is the rate of transferring energy. Joining together $V = W/Q$ and $I = Q/t$ gives the power of any electrical component. Then substituting $V = IR$ gives two more versions you must be able to derive.

Deriving P = VI, P = I²R and P = V²/R

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. From the definition of p.d., the energy transferred when charge Q passes through a component is $W = VQ$. The charge that flows in time t is $Q = It$, so:

$$W = VQ = V(It) \quad \Rightarrow \quad \boxed{W = VIt}$$

2. Power is energy transferred per unit time:

$$P = \frac{W}{t} = \frac{VIt}{t} \quad \Rightarrow \quad \boxed{P = VI}$$

3. From the definition of resistance, $V = IR$. Substitute this for V:

$$P = (IR)\, I \quad \Rightarrow \quad \boxed{P = I^2R}$$

Use this when you know the current, for example for components in series.

4. Or substitute $I = V/R$ for I instead:

$$P = V\left(\frac{V}{R}\right) \quad \Rightarrow \quad \boxed{P = \frac{V^2}{R}}$$

Use this when you know the p.d., for example for components in parallel.

FORMULA SHEET

$$P = VI$$

FORMULA SHEET

$$W = VIt$$

FORMULA SHEET

$$P = I^2R$$

FORMULA SHEET

$$P = \frac{V^2}{R}$$

KEY IDEA

Which resistor gets hotter?

In **series** the current is the same, so $P = I^2R$: the **bigger** resistance dissipates more power. In **parallel** the p.d. is the same, so $P = V^2/R$: the **smaller** resistance dissipates more. That's why a 100 W mains lamp has a lower resistance than a 60 W one.

$P = I^2R$ also explains why electricity is sent across the country at very high voltages. For the same power $P = VI$, a higher voltage means a smaller current, and the energy wasted heating the cables goes as the current *squared*.

WORKED EXAMPLE · MULTI-STEP

A lamp is labelled 12 V, 36 W. (a) Calculate the current in it and its resistance when working normally. (b) Calculate the energy it transfers in 5.0 minutes. (c) A student connects it to a 6.0 V supply and predicts the power will be 9.0 W. Show how she got this value and explain why the real power is larger.

Show solution

(a) From $P = VI$ and then $R = V/I$:

$$I = \frac{P}{V} = \frac{36}{12} = 3.0\ \text{A} \qquad R = \frac{V}{I} = \frac{12}{3.0} = 4.0\ \Omega$$

(b) $W = Pt = 36 \times 300 = 1.1 \times 10^4$ J (10 800 J).

(c) Assuming the resistance stays at 4.0 Ω:

$$P = \frac{V^2}{R} = \frac{6.0^2}{4.0} = 9.0\ \text{W}$$

At 6.0 V the filament is cooler, so its resistance is **less** than 4.0 Ω. With a smaller R and the same V, $P = V^2/R$ is bigger than 9.0 W.

Resistivity

The resistance of a wire depends on its shape as well as what it's made of:

- **Twice the length** is like two identical wires in series, so **twice the resistance**: $R \propto l$.
- **Twice the cross-sectional area** is like two identical wires side by side in parallel, giving the charges twice as many paths, so **half the resistance**: $R \propto 1/A$.

Put together, $R \propto l/A$, and the constant of proportionality is the **resistivity** ρ, a property of the material alone (at a given temperature).

Resistivity

FORMULA SHEET

$$R = \frac{\rho l}{A}$$

ρ: resistivity / Ω m l: length / m A: cross-sectional area / m²

DEFINITION

The **resistivity** of a material is defined by $\rho = \frac{RA}{l}$, where R is the resistance of a sample of length l and cross-sectional area A. Its unit is the ohm metre (Ω m).

Material	Resistivity at room temperature / Ω m
copper	1.7×10^{-8}
constantan (alloy used in CP7)	4.9×10^{-7}
nichrome (heating elements)	1.1×10^{-6}
germanium (semiconductor)	≈ 0.5
silicon (semiconductor)	$\approx 2 \times 10^{3}$
glass	$\approx 10^{10}$ to 10^{14}

PTFE (a plastic)	$\approx 10^{23}$

COMMON MISTAKE

Two classic slips. (1) The micrometer gives the **diameter**; the area is $A = \pi r^2 = \pi d^2/4$. (2) Convert to metres before squaring: $0.40\ \text{mm} = 0.40 \times 10^{-3}\ \text{m}$, and $1\ \text{mm}^2 = 10^{-6}\ \text{m}^2$, not 10^{-3}.

WORKED EXAMPLE

Calculate the resistance of a 1.50 m length of constantan wire of diameter 0.40 mm. Resistivity of constantan = $4.9 \times 10^{-7}\ \Omega\ \text{m}$.

Show solution

$$A = \frac{\pi d^2}{4} = \frac{\pi \times (0.40 \times 10^{-3})^2}{4} = 1.26 \times 10^{-7}\ \text{m}^2$$

$$R = \frac{\rho l}{A} = \frac{4.9 \times 10^{-7} \times 1.50}{1.26 \times 10^{-7}} = 5.8\ \Omega$$

Core Practical 7: resistivity of a wire

CORE PRACTICAL

Determine the electrical resistivity of a material

Tape a length of thin wire (for example constantan) along a metre rule. Connect one end to an ohmmeter (or an ammeter-and-voltmeter circuit) and use a crocodile clip as a sliding contact to select a length l. Measure the resistance R for at least six lengths between about 0.1 m and 1.0 m. Measure the diameter with a micrometer screw gauge.

Since $R = \dfrac{\rho}{A} l$, a graph of R **against** l is a straight line with **gradient** $= \rho/A$. So:

$$\rho = \text{gradient} \times A, \qquad A = \frac{\pi d^2}{4}$$

- **Diameter:** measure it at several points along the wire and in two directions at right angles at each point, then average. Check the micrometer's zero reading first and correct for any zero error. The diameter is the largest source of uncertainty because it gets squared.
- **Heating:** keep the current small and switch off between readings. A warm wire has a higher resistance.

- **Length:** keep the wire straight with no kinks and read the rule at eye level to avoid parallax.
- **Contact resistance** at the clips and the resistance of the leads add the same extra resistance to every reading. That's a **systematic** error: the line has a non-zero intercept, but its gradient, and so ρ, is unaffected. Using the gradient instead of a single reading removes it.
- **Safety:** with short lengths and a high current the wire can get hot enough to burn, so use a small current.

WORKED EXAMPLE

In CP7 a student's graph of R against l has a gradient of 8.4 Ω m⁻¹ and an intercept of 0.30 Ω. Her micrometer readings of the diameter are 0.27, 0.28, 0.27, 0.28 and 0.27 mm. Calculate the resistivity, and suggest what caused the intercept.

Show solution

Mean diameter $= 0.274\ \text{mm} = 2.74 \times 10^{-4}\ \text{m}$, so

$$A = \frac{\pi(2.74 \times 10^{-4})^2}{4} = 5.90 \times 10^{-8}\ \text{m}^2$$

$$\rho = \text{gradient} \times A = 8.4 \times 5.90 \times 10^{-8} = 5.0 \times 10^{-7}\ \Omega\ \text{m}$$

The 0.30 Ω intercept means every reading includes an extra resistance that doesn't depend on l: the contact resistance of the crocodile clips plus the resistance of the connecting leads (or an ohmmeter zero error). It doesn't affect the gradient, so ρ is still valid.

Resistance, temperature and light

The spec asks you to explain changes in resistance using a simple model with two ingredients:

1. the **number of conduction electrons** per unit volume, n;
2. how often those electrons collide with the **vibrating lattice ions**, which limits their drift velocity v.

Then use $I = nqvA$: anything that reduces n or v for a given p.d. reduces the current, so it increases the resistance.

Metal: resistance rises with temperature

The number of conduction electrons is already huge and doesn't change. As the temperature rises the lattice ions vibrate with larger amplitude,

NTC thermistor: resistance falls with temperature

A thermistor is a semiconductor with few conduction electrons. As it warms up, more

so the electrons collide with them more often. The drift velocity for a given p.d. decreases, so the current decreases and the resistance **increases**.

electrons gain enough energy to break free and conduct, so n increases a lot. The lattice vibrations increase too, but the increase in n has a much bigger effect, so the resistance **decreases**. NTC means negative temperature coefficient.

LDR: resistance falls in brighter light

A light-dependent resistor is also a semiconductor. Photons of light give bound electrons enough energy to become conduction electrons. More light releases more electrons, so n increases and the resistance **decreases**: from around a megaohm in the dark to a few hundred ohms in bright light.

EXAM TIP

For a thermistor, give both halves of the argument: "more conduction electrons are released, **and** this outweighs the effect of the increased lattice vibrations". For a metal, say the number of conduction electrons stays the same.

Quick checks

QUICK CHECK

A current of 0.40 A flows for 2.0 minutes. How much charge passes a point?

- 0.80 C
- 5.0 C
- 48 C
- 300 C

$\Delta Q = I\Delta t = 0.40 \times 120 = 48$ C. The first option forgets to convert minutes to seconds.

QUICK CHECK

A copper wire has a thin section with half the cross-sectional area of the rest. How does the drift velocity in the thin section compare with that in the thick section?

- Half as big
- The same
- Twice as big
- Four times as big

The sections are in series, so the current is the same in both (charge is conserved), and n and q are the same. With $I = nqvA$ fixed, halving A doubles v.

QUICK CHECK

A wire is stretched until it is twice as long. Its volume stays the same. What happens to its resistance?

- It doubles.
- It becomes four times as big.
- It stays the same.
- It halves.

Same volume, twice the length, so the area halves. $R = \rho l/A$ has l doubled *and* A halved, so R goes up by $2 \times 2 = 4$.

QUICK CHECK

At one point on the I–V graph of a component, $V = 4.0\ \text{V}$ and $I = 0.20\ \text{A}$. The tangent to the curve there has a gradient of $0.10\ \text{A V}^{-1}$. What is the resistance of the component at this point?

- 0.050 Ω
- 0.80 Ω
- 10 Ω
- 20 Ω

Resistance is defined as $R = V/I = 4.0/0.20 = 20\ \Omega$. The 10 Ω option is 1/gradient, which is not the resistance unless the graph is a straight line through the origin.

QUICK CHECK

Why does the resistance of an NTC thermistor decrease as its temperature increases?

- The lattice ions vibrate less, so there are fewer collisions.
- More conduction electrons are released, and this outweighs the effect of the increased lattice vibrations.
- The electrons move faster, so there are fewer collisions.
- The thermistor expands, so its cross-sectional area increases.

Heating makes the lattice vibrate *more*, which on its own would raise the resistance (as in a metal). In a semiconductor the number of conduction electrons n increases enormously, which wins.

QUICK CHECK

A 60 W lamp and a 100 W lamp are both designed for the 230 V mains. Which statement is correct when they are working normally?

- The 100 W lamp has the larger resistance.
- The 60 W lamp has the larger resistance.
- They have the same resistance because the p.d. is the same.
- The 60 W lamp carries the larger current.

$R = V^2/P$. Same V, so a smaller power means a larger resistance: $230^2/60 = 880\ \Omega$ against $230^2/100 = 530\ \Omega$. The 100 W lamp carries more current ($I = P/V$).